

①

	S ₁	S ₂	S ₃	S ₄
S ₁	0	2	4	3
S ₂		0	5	4
S ₃			0	6
S ₄				0

* La menor distancia es de S₁, S₂ (2)

$$S_5 = \{S_1, S_2\}$$

$$d(S_5, S_4) = \frac{d(S_1, S_4) + d(S_2, S_4)}{2} = \frac{3+4}{2} = 3.5$$

$$d(S_5, S_3) = \frac{d(S_1, S_3) + d(S_2, S_3)}{2} = \frac{4+5}{2} = 4.5$$

$$d(S_3, S_4) = 6$$

	S ₅	S ₃	S ₄
S ₅	0	4.5	3.5
S ₃		0	6
S ₄			0

* La menor distancia es de S₅, S₄ (3.5)

$$S_6 = \{S_4, S_5\}$$

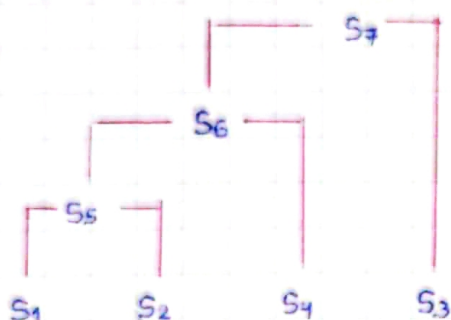
$$d(S_6, S_3) = \frac{n_i d(S_5, S_3) + n_j d(S_4, S_3)}{n_i + n_j} = \frac{2(4.5) + 1(6)}{2+1} = \frac{9+6}{3} = 5$$

	S ₆	S ₃
S ₆	0	5
S ₃		0

* La menor distancia es de S₆, S₃ (5)

$$S_7 = \{S_6, S_3\}$$

* El árbol guía es el siguiente



② Alineamiento múltiple de secuencias

* S₁, S₂

S₁ = A T T X B B C A C C A

S₂ = A T T T B B X A C C A

* S₁, S₄

S₁ = A T T X B B X C A C C A X

S₄ = A T T X X X C C A C C A C

* S₂, S₄

S₂ = A T T T B B X X A C C A X

S₄ = A T T X X X C C A C C A C

* S₁, S₃

S₁ = A T T X B B X C X C C A X

S₃ = X X T X B B T T X C C A X

* S₂, S₃

S₂ = A T T T B B X X A C C A X

S₃ = X X T X B B T T X C C A X

* S₄, S₃

S₄ = A T T X X X C C A C C A C

S₃ = X X T X B B T T X C C A X

S₁ = A T T X B B X C X C C A X

S₂ = A T T T B B X X A C C A X

S₃ = X X T X B B T T X C C A X

S₄ = A T T X X X C C A C C A C

* Por lo tanto

S₁ = A T T - B B - C - C C A -

S₂ = A T T T B B - - A C C A -

S₃ = - - T T B B T T - C C A -

S₄ = A T T T - - C C A C C A C //